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CSC 6303 Week 6

File system basic concepts and styles

Systems and Languages Survey - Dr. Paulo Fernandes

Presentation Agenda

Week 6

- File Systems
  a. Principles
  b. Kinds and examples
- Dealing with file systems
  a. Drives, folders and files
  b. Links and aliases
  c. Virtual drives, FTP, and the cloud
- This Week's tasks

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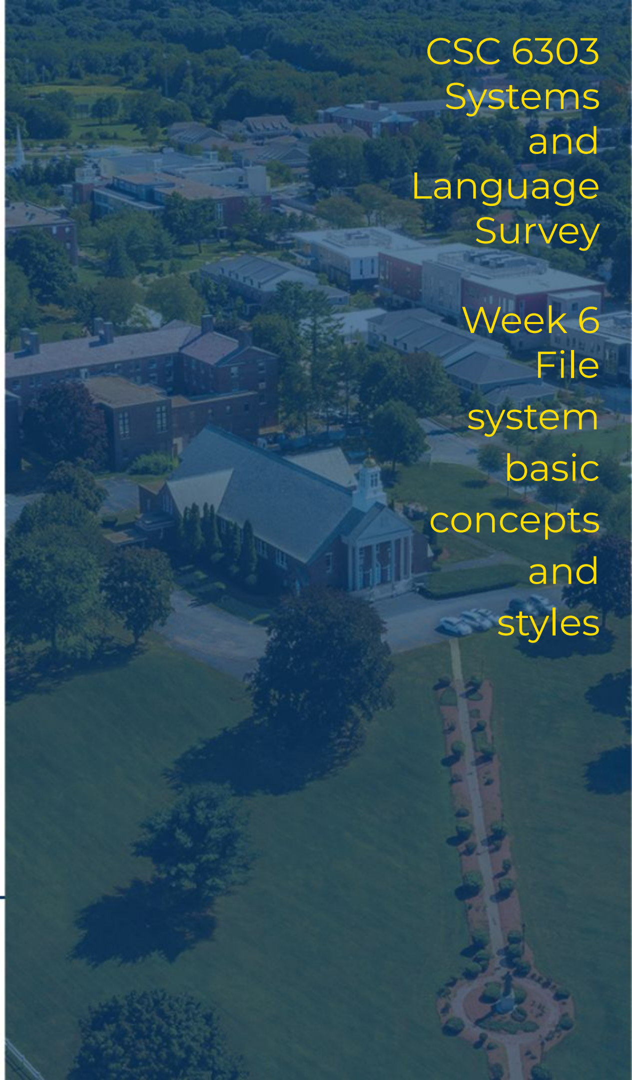


File Systems

Storing large amounts of information is easy, but it becomes quite complicated when you also need to retrieve the stored information.

File systems are present since computers exist. They have changed considerably according to the technology. Operating Systems play an important role in file system characteristics, but the tree structure is generally accepted as the meta model for all file systems.

There are families of file systems, and currently some options became standards, and we will see it by:

- Principles
- Kinds and examples

File Systems through time

Search the data

The first file systems (50's) were stored in magnetic tapes, and because of that the information was stored assuming a sequential search for the information you are looking for.

Shortly after (60's) magnetic hard disks were used to store files, thus allowing a much faster access in a tree-like structure. The floppy disks followed the same idea and that was the standard solution until the turn of the century.

Pick the [disk]
Pick the [track]
Search the data

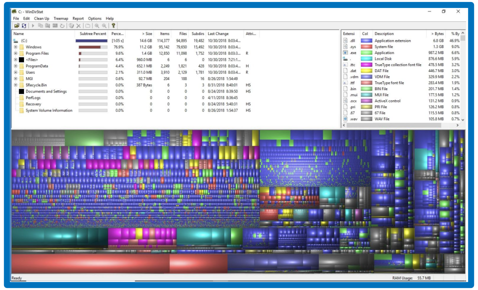
Solid state memory allowed, first with flash drives, then with SSDs the replacements of hard drives for a much faster medium with indexed access.

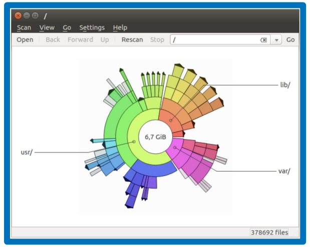
Pick the [data]

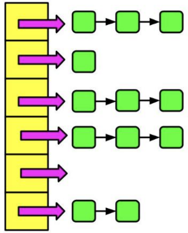
Wikipedia: Computer data storage.

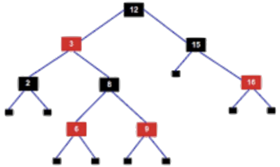
File System Principles

- Space management
  - memory space is composed by sectors, each item (files, folders, etc.) is stored in one or more sectors, those sectors do not need to be contiguous;
  - given the item name (key) the file system fetch the sector(s) where the item is stored;
  - choices of storage to make it easier to retrieve;
- Item names as keys
  - acceptable name format;
  - indexing structure (hash, red & black trees, etc);
- Security
  - system areas, personal areas, access rights.









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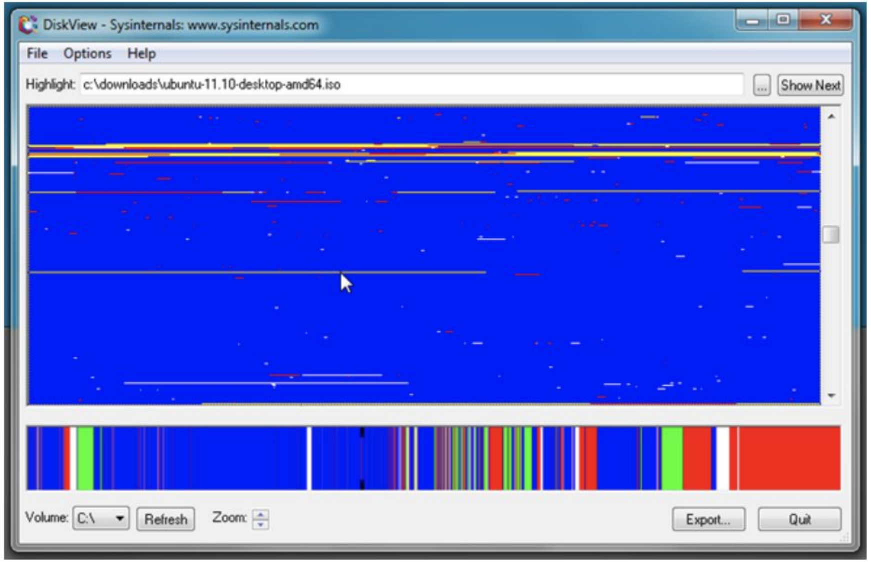
Wikipedia: File System.

Space Management

- A memory device is splitted in a control part to index the usable parts, which are divided in sectors;
- Each time an item (folder or file) is created, the file system chooses, according to an criteria, which sector will be allocated to the item;
  - the criteria can be first available, random choice, or best fit (aiming or avoiding contiguity - or other ones considering the physical device).
- Each time an item needs to be accessed, its sectors are read;
- Each time an item is deleted this is done only in the control part.



Text: Hard disk drives and Solid-state drives.

Item name conventions

Depending on the file system there are very specific rules for item names, but there are some best practices to consider if you want to change file systems:

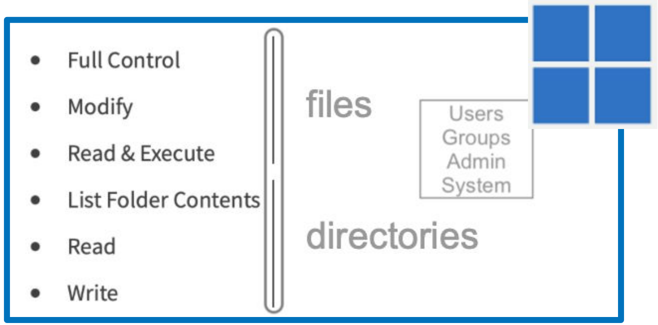
- Always use the same case:
  - old systems convert always to uppercase (not case-preserving ones);
  - some systems are case-preserving, but case insensitive, "*File.txt*", "*file.TXT*", and "*file.txt*" are the same file;
  - some systems are case sensitive, "*FILE.txt*" and "*file.txt*" are different;
- Avoid starting with something that is not a letter;
- Use only letters, digits, and the symbols:
  - "." used to separate parts, including the extension (after the last ".");
  - "_" used to separate parts (avoid camelBack in item names);
  - " " blank space can an additional complication while using it in scripts.

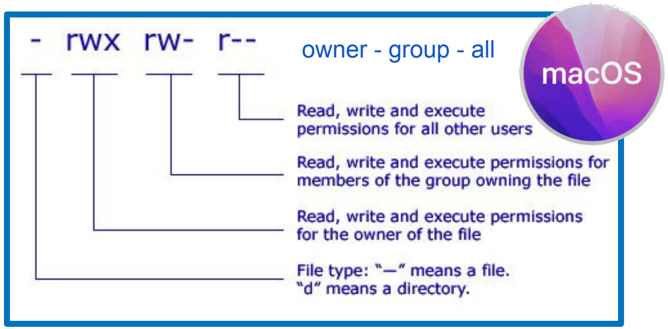
Wikipedia: Filename.

Security - Access Control List - Permissions

Depending on the file system there different level of security, and according to the OS principles this can be dealt very differently.

- Old windows based systems didn't differentiate system files and folders from user ones, while Unix-based systems always differentiate system files and folders from user ones;
  - Current systems have strict policies about security and privacy, some that can be toggled off, others don't.





Text: [Windows File and Folder]() - [Unix-Linux-MacOS]().

File Systems Kinds

According to media:

- Tape file systems;
- Disk file systems;
- Flash file systems.

According to usage:

- Standard file systems;
- Database file systems;
- Transactional file systems;
- Network file systems.

According to OS features:

- Unix-like operating systems oriented file systems;
- Old Windows compatible file systems;
- New Windows compatible file systems.

Wikipedia: File System.

Examples - More common File Systems

- exFAT
  - An extension of populars FAT and FAT32, readable by most OSs and devices;
- NTFS
  - The first modern file system adopted by Microsoft for Windows servers, readable by most OSs;
- EXT4
  - The latest version of the Linux-based file systems (after EXT2 and EXT3), readable by most Unix-like OSs;
- APFS
  - The Apple's file system, efficient and configurable, usually readable by Apple and compatibility modules on Linux.

Text: Which File System should I use?

Dealing with file systems

Most file systems particularities are encapsulated by the OS view of files and folders.

Practically, the use of the file system is mostly affected by the Operating System format to view and access the file system items.

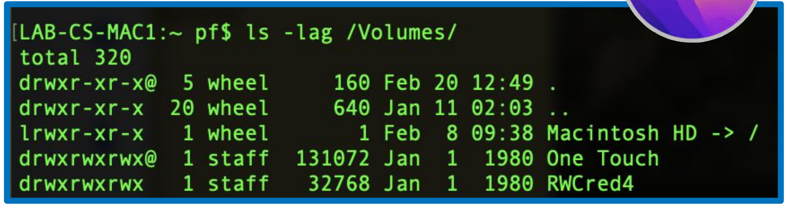
This is perceived by users in terms of:

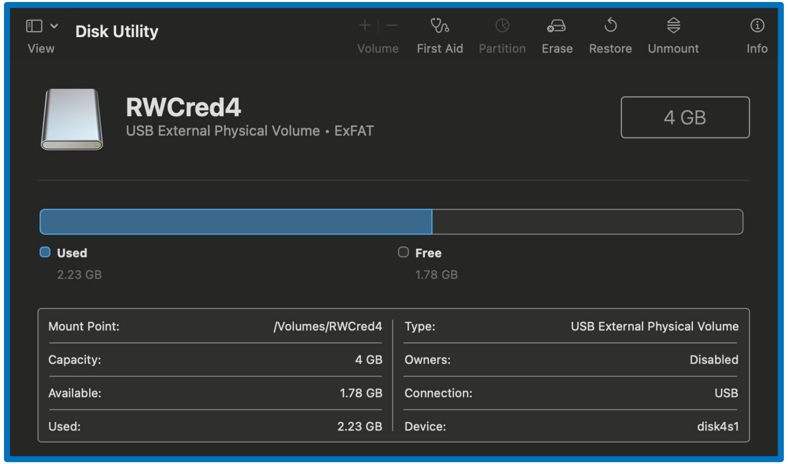
- Drives, folders and files
- Links and aliases
- Virtual drives, FTP, and the cloud

Drives in Unix-like systems

The definition of drives is masked on Unix-like systems as the whole file system is mounted on a single root (***/***). The main OS volume is build at the root and all other devices are physically associated within a slot in the devices folder (***/dev/***), and mounted at the volumes folder (***/Volumes/***).

For example, the ExFAT flash drive volume named RWCred4 is mounted over ***/dev/disk4s1*** at ***/Volumes/RWCred4***.



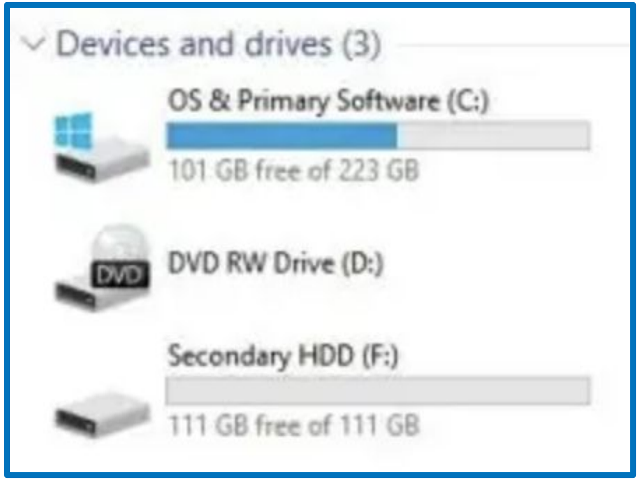


Text: Mount & Unmount Drives from CLI Mac OS.

Drives in Windows systems

Windows 11

The definition of drives is explicit on Windows systems as the file system is mounted assigning a capital letter followed by colon (for example ***F:***) to each mounted drive, including removable media (floppies, CDs, DVDs, etc). Historically, ***A:*** and ***B:*** are used for floppy drives, which are rarely used, so, basically now only letters from ***C:*** are alphabetically assigned.



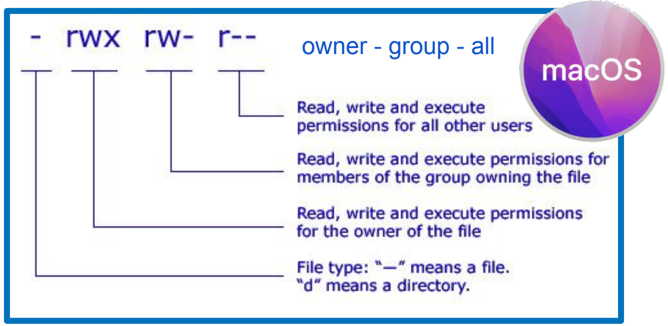
Note that removable drives, as network devices, also receive letters descriptors.

Text: How to see all drives available on the computer.

Folders and Files in Unix-like systems

A folder or a file name in Mac OS, depending on how the drive was formatted, may be case-sensitive or case-preserving. It may contain blank spaces (refer as "**** " in CLI, e.g. "*My File.txt*" is written ***My\\ File.txt***), but it may be or not properly dealt in other Apps. The access to files and folders is protected with mandatory OS restrictions, and then using the *owner-group-all read-write-execute* permission system. The access through ***finder*** app is configurable with hide/show toggles.

Basic CLI commands have the same rights as the ***finder*** with full visibility, plus the super user option (***su*** CLI command).



Video: macOS File Storage System.

Folders and Files in Windows systems

Essentially, the file and folders are handled similarly in Windows except for a few differences on how they are explicitly referred in CLI.

- In Unix-like we use a **/** to separate folders, in Windows we use a **** ;
- In Unix-like a name with blank space replace blanks by "**** ", in Windows the full name must be quotation, e.g.: ***"My file.txt"***.

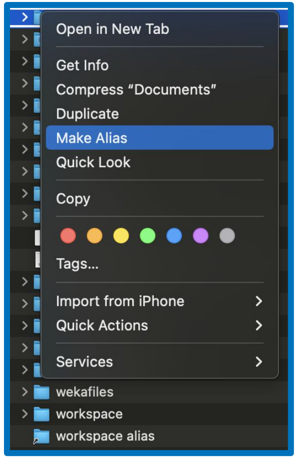
Many concepts are similar, as **.** and **..** are used to refer to the current directory and the directory above.

In windows ***C:text.txt*** refers to the file ***text.txt*** at the current directory of drive **C:**, while ***C:\text.txt*** refers to the file ***text.txt*** at the root folder of drive **C:**.

In Windows security of files and folders is provided using the ACL concept.

Video: [Linux vs Windows File System Compared!]

Links and Aliases in Unix-like systems



In Mac OS and other Unix-like systems it is possible to create shortcuts to provide quick access and encapsulated access to files and folders.

- **Aliases**, available through ***Finder***, not CLI, are simple shortcuts to just jump from one location to another using the GUI;
- **Symbolic links**, available through CLI only, behave similarly to aliases, but they do not update as the target moves
- **Hard links**, available through CLI only, create a bound between the original file and the linked one, so you cannot delete the original while the link to it exists.

```
[LAB-CS-MAC1:~ pf$ ls -lag workspace*
-rw-r--r--@ 1 staff  800 Feb 20 14:46 workspace alias

workspace:
total 16
drwx------   4 staff   128 Feb 18 18:19 .
drwxr-xr-x+ 57 staff  1824 Feb 20 14:49 ..
-rw-r--r--@  1 staff  6148 Feb 18 18:19 .DS_Store
drwx------   5 staff   160 Feb 18 18:19 Unitex-GramLab
LAB-CS-MAC1:~ pf$
```

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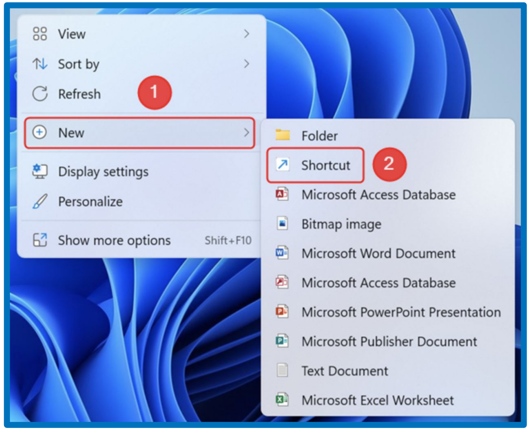
Text: [Shortcuts in Mac OS](#).

Links and Aliases in Windows systems

In Windows systems it is also possible to create shortcuts to provide quick access and encapsulated access to files and folders.

- Mac OS aliases are called in Windows **shortcuts** and they behave quite similarly (they are available using the right-click menu);



- **Symbolic links** are also available in Windows using CLI through command line prompt or using PowerShell.

```
New-Item -ItemType SymbolicLink -Path "Link" -Target "Target"
```


Text: Shortcuts and Symbolic links in Windows.

Virtual Drives

A common usage in Operating Systems is the usage of virtual drives, i.e., drives that are physical, but just a format that follows a given standard and is dealt by the OS as a connected device.

The more common use of such virtual disks is done using the standard ISO-9660 defined by the International Standards Organisation. Such formats are known as an ***ISO image***, and they are commonly employed to replace a CD or DVD disk, or even a flash drive. In both modern Mac OS and Windows the ISO images are automatically handled by the OS by either double clicking an ISO image file, or right-clicking and choosing mount. In Mac OS is also frequent to find a DMG format (Apple Disk iMaGe) with similar behavior.

Creating a disk image in Mac OS or in Windows.

FTP - File Transfer Protocol

One of the first service available in the Internet is the File Transfer Protocol (FTP), which is a way to connect with another machine and exchange files and folders from the file system in your machine and the file system in the other machine.

```
SFTP(1)                   General Commands Manual                   SFTP(1)

NAME
     sftp - OpenSSH secure file transfer

SYNOPSIS
     sftp [-46AaCfNpqrv] [-B buffer_size] [-b batchfile] [-c cipher]
          [-D sftp_server_path] [-F ssh_config] [-i identity_file]
          [-J destination] [-l limit] [-o ssh_option] [-P port]
          [-R num_requests] [-S program] [-s subsystem | sftp_server]
          destination

DESCRIPTION
     sftp is a file transfer program, similar to ftp(1), which performs all
     operations over an encrypted ssh(1) transport.  It may also use many
     features of ssh, such as public key authentication and compression.

     The destination may be specified either as [user@]host[:path] or as a URI
     in the form sftp://[user@]host[:port][/path].
:
```

The current version of FTP is available using sftp command (Secure FTP).

Video: How to use SFTP Commands.

FTP - File Transfer Protocol and The Cloud

Using FTP, the file system differences are ignored (except for file names restrictions). It can be done using both CLI or GUI, or dedicated Apps, like Transmit for Mac OS or WinSCP for Windows.

Another possible way to perform file transfer is through a FTP web service. In fact, cloud storage services, as Google drive for example, usually offer their own version of visualization, editing, or transfer of files and folders from your machine and the cloud repository.

Video: [WinSCP Tutorial].

This Week's tasks

- Post discussion
- Programming project #6

Tasks

- Comment on **if you realized that there were different file systems for portable memory devices.**
- Create a Python program **that discovers the current OS, then asks the user a file name with path, and converts it according to the current OS.**

Post Discussion - Week 6 - In-class exercises

Likely in your previous experiences with portable devices (external drives, flash drive, etc) you may not realized that there were different file systems for them.

- Have you plugged in a flash drive in a smart TV?
- Have you transported files from a machine to another?
  - In such kind of experiences were you aware that the machines and portable memory device might had different file systems?
- And now that you are aware, how this would impact your behavior?

Your task:

- Post your discussion in the message board until this Saturday;
- Reply posts of your colleagues in the message board until next Wednesday.

This task counts towards the In-Class Exercises grade.

Progr. Project #6 - Week 6 - P#6

Create a Python program that discovers the current OS and asks the user a file name with path and converts it according to the system (basically converting drive, folder separators and blank spaces).

- Your program should use the packages ***platform*** and ***os*** (see the example here).
- Once your program is aware of the platform and OS, it should ask the user a file name with the path (it can be entered according to any format: MacOS or Windows), and your program should convert it to the right format according to the platform and OS in which the program is running (MacOS and Windows).

`platform` — Access to underlying platform's identifying data

- Cross Platform
- Java Platform
- Windows Platform
- macOS Platform
- Unix Platforms
- Linux Platforms

`os` — Miscellaneous operating system interfaces

- File Names, Command Line Arguments, and Environment Variables
- Python UTF-8 Mode
- Process Parameters
- File Object Creation

Progr. Project #6 - Week 6 - P#6

- You must submit the Python code in a single file.
- If you need to add any explanation or messages to me, please make it in the code remarks.
- The quality of your code remarks is not relevant for the evaluation, but make sure you write your name in the remarks, as well as to which system (version included) your code was tested on.
  - It is not necessary, but it is a good practice to properly document your code.

Your task:

- Save your file and submit it in the appropriate Canvas until Next Wednesday.

This assignment counts towards the Coding Projects grade.

Week 6 of CSC 6303

- **Post in the discussions until Saturday, reply until next Wednesday**
- **Do the coding project until next Wednesday**

Next Week - OS examples: Linux, MacOS, and Windows